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How many zones should an electricity market have? A cross-country perspective on bidding zone design

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ABSTRACT

The configuration of bidding zones has become a central issue in the ongoing debate on electricity market design. This paper critically analyses the effectiveness and limitations of zonal pricing through a comparative analysis of Italy, Norway, Sweden and Australia-with zonal systems-and of Texas and California, which moved from zonal to nodal regimes. We analyze the institutional, technical and socio-economic factors that shape these divergent trajectories, highlighting how national authority in the European context and energy system characteristics influence market structure. In zonal markets, locational pricing applies on the supply side, while demand-side application varies. By examining zoning processes and price convergence, we identify both shared patterns and context-specific dynamics. While zonal pricing can enhance transparency and investment signals, its effectiveness depends on periodic bidding zones reviews. The findings support a flexible approach to bidding zone design, aligned with the objectives of decarbonization, market integration, and system reliability.

1. Introduction

Electricity markets worldwide are increasingly challenged by network congestion, driven by growing electricity demand, expanding renewable generation and transmission infrastructure constraints. Ensuring grid reliability while facilitating efficient market functioning—especially under conditions of climate variability—has placed congestion management at the center of power system design, as the recent blackout in Spain and Portugal on April 28, 2025 has highlighted (ENTSO-E, 2025).

In this context, increasing the spatial granularity of price signals has been identified as a key means of enhancing market efficiency. Two principal approaches have therefore emerged. The first involves a shift from a national market to a zonal configuration, by partitioning the transmission network into limited predefined geographic areas or "bidding zones" (BZs),[2] in which electricity prices are assumed to be uniform. The second approach entails a more granular design, in which the market evolves beyond a zonal framework—often through an intermediate zonal configuration, as adopted in several U.S. markets—toward a fully nodal system, where prices are determined at each network node. Although nodal pricing theoretically provides the most efficient representation of network constraints, in practice, zonal pricing has become the dominant approach across the European Union, reflecting a balance between market efficiency, system feasibility, and institutional practicality.

Today, the zonal pricing model is increasingly under scrutiny. Despite its simplicity and political acceptability, it seems increasingly evident that a regular review of zones is necessary for its effectiveness. Intra-zone transmission constraints often require costly redispatch actions by the national Transmission System Operators (TSOs) after the market has cleared, resulting in inefficiencies, distorted price signals and increased system costs, undermining the basic assumption of homogeneity within zones. These trends raise questions about the adequacy of the current zonal configuration and its ability to cope with increasing intra-zonal bottlenecks and ensure economically efficient energy flows.

This paper evaluates the effectiveness of Europe's zonal pricing framework, comparing different experiences. Despite the centrality of this issue, the academic literature still lacks a comprehensive analysis of

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[2] There is no standard size for market zones, but they vary enormously in terms of territory, encompassing extremes such as an entire zone reflecting national borders, such as Germany, and smaller countries divided into several zones, such as Italy.

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how the number of bidding zones influences market performance. This paper seeks to address this gap by providing a comparative assessment of different zonal market structures, illustrating how adjustments in zonal granularity affect price formation and market efficiency.

The remainder of the paper is organized as follows. Section 2 reviews the academic literature on methodologies for defining bidding zones, highlighting both theoretical foundations and empirical approaches. Section 3 outlines the legal and institutional framework governing the bidding-zone reconfiguration process within the European context. Section 4 presents a comparative analysis of zonal pricing systems across a selection of markets that have experienced a zonal structure in the past (Texas and California) and those that currently have it (Australia, Norway, Sweden and Italy) illustrating both convergences and divergences in design and evolution. Section 5 draws lessons from these case studies to identify best practices and policy-relevant insights for the reconfiguration of zonal electricity markets. Finally, the paper concludes by discussing broader implications for ongoing electricity market reforms in Europe and beyond.

2. Zonal pricing in the European electricity market

In Europe, the zonal market design was implemented during the 1990s and early 2000s as part of the transition from vertically integrated monopolies to competitive wholesale electricity markets. Although the concept of nodal pricing was already known at the time—with some markets, such as New Zealand (1997) and the Pennsylvania–New Jersey–Maryland (PJM) interconnection (1998), having adopted nodal pricing—European countries and neighbouring nations opted for zonal pricing instead (Eicke and Schittekatte, 2022).

Although the zonal system is in theory recognized as a natural starting point, reflecting the initial approach in US markets which then quickly moved towards nodal prices,[3] European zonal systems have shown resistance to this change. The decision to implement zonal pricing in the European electricity market was primarily the result of an historical and political compromise between efficiency and perceived fairness to consumers and investors. The integration of the European electricity market required a substantial regulatory and institutional coordination effort that could accommodate national sovereignty and regulatory diversity among Member States. A nodal approach would have necessitated the creation of a centralized authority capable of coordinating market design reforms across national jurisdictions, thereby superseding the control of national regulatory agencies. At the time, such centralization was not politically feasible. Given these constraints, the European market design favoured a simplified pricing model that respected national borders (Sarfati et al., 2019), thereby minimizing political and economic frictions and enabling a decentralized integration of electricity markets through market coupling. Zonal pricing, therefore, emerged as the pragmatic solution,[4] despite the theoretical advantages often attributed to nodal pricing models.

More recent proposals (Neuhoff et al., 2025) outline coordination frameworks that could allow member states to adopt different levels of market granularity within a harmonized European setting. While these designs are conceptually promising, they remain largely untested in practice; the present analysis therefore concentrates on the existing zonal architectures underpinning the current European market-coupling framework.

The European approach entails the application of a uniform price within each bidding zone, thereby promoting greater transparency through a single price signal across a broad geographic area and limiting the market power that individual actors can exert (Dobos et al., 2025). In a multinational context, this framework also enhances market liquidity and facilitates cross-border electricity trade—an essential element for preventing bottlenecks and minimizing congestion costs at interconnection points. However, the use of a uniform electricity price also presents significant drawbacks (NEMO Committee, 2024). It fails to provide local incentives to adapt to the volatility introduced by increasing shares of renewable energy sources. Furthermore, it can result in notable welfare losses stemming from the constraint of uniform pricing, and it incurs high redispatching costs to address internal congestion within each zone.

By comparison, PJM, one of the largest wholesale electricity markets in the US, is a well-established example of the implementation of the nodal structure. However, both geographic and electrical scales of the European electricity market, which currently coordinates 61 BZs spread across several Member States (as shown in Fig. 1), are significantly larger than those of PJM, which spans 13 federal states and the District of Columbia.[5] Moreover, these 61 zones vary in their configuration: some align with national borders, such as France and Spain, simplifying governance and regulatory control; Germany and Luxembourg are united in the same market area; others are sub-national—as Italy, Sweden and Norway— which are subdivided into several zones to better reflect structural transmission constraints and regional differences in supply and demand. This difference in spatial extent, coupled with greater regulatory and infrastructural heterogeneity, poses structural limits to the uniform application of a continent-wide nodal regime. However, over the years, some EU Member States have proposed the implementation of a nodal system. Poland, for example, had planned to implement a nodal pricing system in its internal market in 2015, but the project was abandoned. Although several studies, such as Bjørndal et al. (2018), found that Poland could significantly reduce redispatch costs, the proposal was hindered by neighbouring countries, like Germany, which feared that this reconfiguration would disrupt the integrated zonal market structure, complicating the management of cross-border flows and undermining harmonization at the European level (Frontier Economics and Consentec, 2024). Consequently, although the US experience represents a useful technical benchmark, its full replicability in the European context appears limited, suggesting instead a differentiated and gradual approach.

A final instructive example is Great Britain, where multiple regions are aggregated into a single price zone. In the 1990s, amid widespread privatization under Prime Minister Thatcher's government, England and Wales transitioned from the integrated CEGB[6] model to a competitive market, while Scotland retained two vertically integrated utilities, maintaining a distinct though partially deregulated structure (Grubb and Newbery, 2018; Davidson and Odubiyi, 2005). It was only in 2005, with the introduction of the British Electricity Trading and Transmission Arrangements (BETTA), that Scotland was formally integrated into a single British market, despite persistent congestion on the Anglo-Scottish border. In the meantime, Northern Ireland remained institutionally separate from the British market and subsequently developed a unified wholesale market with the Republic of Ireland—the Single Electricity Market (SEM)—which became operational in 2007, creating a single unified electricity trading system across the island of Ireland (CER, 2005).

[3] It is worth noting, however, that in some markets national pricing has been replaced directly by a nodal system, as in the case of Ontario from 2025 (IESO, 2025) and Alberta from 2027 (AESO, 2025).

[4] In its simpler application, the zonal system provides for an implementation time of about 18 months, compared to a minimum of five years for nodal, which makes it preferable in case of necessary reforms (Ove Arup and Partners Limited, 2024).

[5] In terms of size, the ENTSO-E area spans over 4 million km^2 and coordinates more than 1200 GW of installed capacity with peak loads exceeding 400 GW, across 35 countries and 61 bidding zones; in contrast, PJM covers approximately 820,000 km^2, with a peak load of around 150 GW and an installed capacity of about 180 GW.

[6] Central Electricity Generating Board.

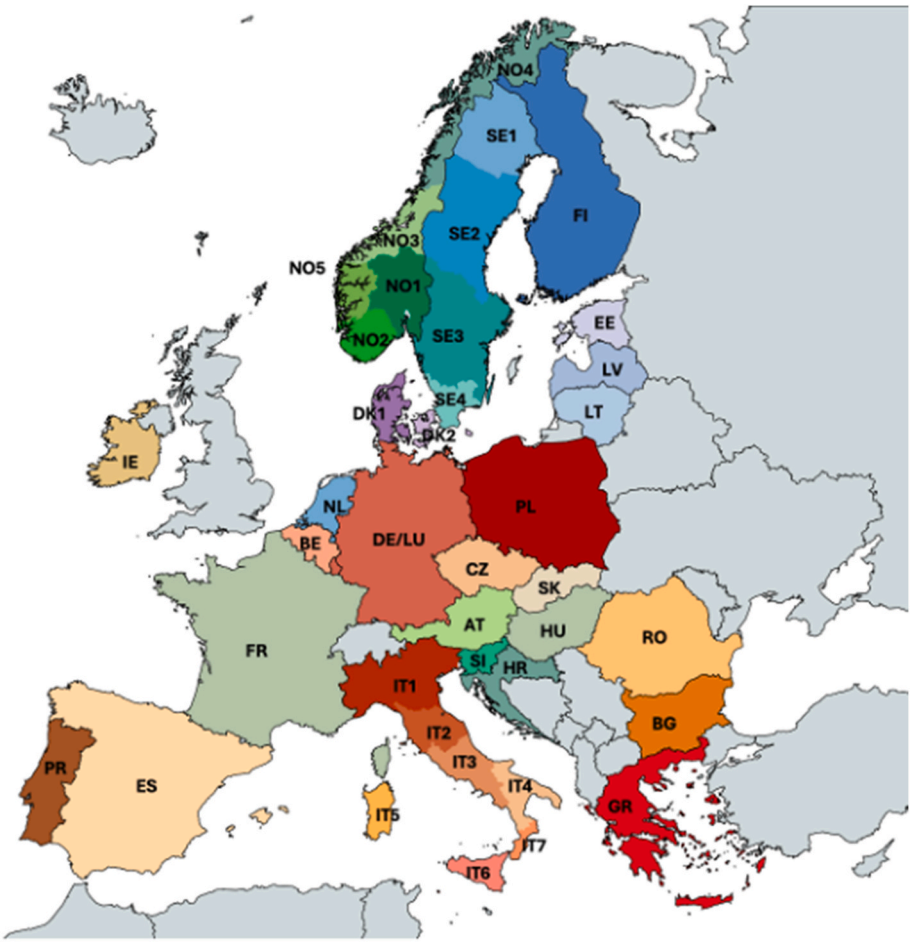

Fig. 1. Main Bidding Zones in the European Electricity Market.
Source: Authors' elaboration using MapChart software (https://www.mapchart.net).
Note: Zone boundaries are based on publicly available ENTSO-E and ACER bidding zone definitions (See: https://bit.ly/ENTSO-E_Bidding_Zone_Configuration). Official ENTSO-E zone names are used. Not all 61 zones are shown.

3. Literature review on defining bidding zones

The European Target model for the internal electricity market (IEM) provides for electricity markets—both day-ahead and intra-day—structured around a zonal representation of the European electricity system. This zonal logic also extends to the balancing market, where Frequency Recovery Reserves (FRR), in particular within the Europe-wide Manually Activated Reserves Initiative (MARI) platform, are generally purchased and settled at the supply-zone level. In these frameworks, BZs play a crucial role in market design and must be established in a way that optimizes congestion management and enhances overall market efficiency, as outlined by the guideline EU Regulation 2015/1222 on Capacity Allocation and Congestion Management (CACM). As a result, defining BZ boundaries is a key aspect of the IEM and necessitates detailed analysis.

Several studies have analyzed evaluation criteria for defining BZs. The CACM framework sets fundamental guidelines for the definition of zones, considering factors such as economic welfare, market liquidity, competition, network structure and topology, planned network reinforcement, and redispatch costs. However, the European Network of Transmission System Operators for Electricity (ENTSO-E) places greater emphasis on market concentration, liquidity, and price signals (ENTSO-E, 2024). Building upon this foundation, some studies have proposed additional criteria for BZ evaluation. Supponen (2011) suggests including the direction of wind power flows, while Breuer and Moser (2014) highlights the importance of generation costs, network security constraints, and market power potential. Additionally, Sarfati et al. (2015) introduces five key indicators for assessing the impact of different BZ configurations.[7] Similarly, Bemš et al. (2016) identifies criteria that prioritize social welfare.[8] Brouhard et al. (2020) proposes a method that combines sophisticated static grid models with dynamic market simulations, enabling a comprehensive analysis of market efficiency. Lastly, Griffone et al. (2019) defines ten essential requirements for an effective zonal configuration based on quantitative and objective parameters. However, few existing studies provide a direct comparison between zonal and nodal markets,[9] and much of the literature refers to the case of the market in Texas rather than the European context.

4. Legal framework of the bidding zone reconfiguration process

In the European electricity market, the Clean Energy Package establishes that the borders of bidding zones must reflect long-term structural congestion, unless these are mitigated or remain locally confined. Each TSO is required to assess and report on the efficiency of its market's zonal configuration. This principle highlights the need to

[7] Commercial Exchanges Evolution, Price Convergence, Price Divergence, Social Welfare Evolution, and Loop Flows.

[8] Including Congestion Rent, Remedial Actions, Marginal Price Differences, Price Volatility, Transition Costs, and Social Welfare.

[9] See, among the contributions on the topic, the following studies: Zarnikau et al. (2014), Triolo and Wolak (2011, 2022).

reconcile efficient congestion management with the broader objectives of market integration and subsidiarity.

In assessing zonal configurations, market liquidity is a key element. The literature generally argues that larger zones improve liquidity due to a greater number of participants. Studies such as Hary (2018), ACER (2014), and Laur and Küpper (2020) show that, by contrast, smaller zones tend to have lower liquidity, making it harder to hedge risks and increasing market power issues, resulting in welfare losses. However, smaller zones allow for more precise congestion management by converting unscheduled flows into market-regulated transactions (ČEPS, MAVIR, PSE Operator and SEPS, 2012). Some studies, such as Newbery (2002) and Hirth (2006), emphasize that larger zones shift congestion management to their borders, while smaller ones internalize physical network constraints without necessarily reducing liquidity. Meanwhile, Consentec for EEX & EPEX SPOT (2015) argue that creating smaller zones entails higher transition and implementation costs, affecting bidding behaviour and widening bid-ask spreads. Liquidity is also critical in intraday markets, which have historically been harder to implement in nodal pricing systems, such as US nodal markets (Herrero et al., 2016). Recent efforts, such as the introduction of intraday auctions in PJM, aim to improve short-term flexibility, though they may also introduce inefficiencies if producers remain uncertain about dispatch outcomes. Moreover, low liquidity can distort the value of Financial Transmission Rights (FTRs), reducing their effectiveness as hedging instruments. In nodal markets, intraday liquidity has historically been more limited, although recent reforms—such as PJM's intraday auctions—have sought to improve it.

In 2018, the first European bidding-zone review process was initiated but remains ongoing due to significant changes reflecting political resistance and lack of consensus among TSOs. This process highlights a fundamental tension between what may be optimal for the efficiency of an integrated European electricity market and the principle of subsidiarity, which prioritizes national autonomy. Initially, on October 5, 2019, all European TSOs submitted a zonal proposal to the relevant regulatory authorities for approval. However, this proposal did not include alternative bidding-zone configurations for a large part of the European territory. Subsequently, by April 7, 2020, the TSOs submitted an updated version of the proposal, which was then sent by the regulators to ACER for a final decision.

On November 24, 2020, ACER issued its first decision in terms of revising the zonal configuration, adopting the methodology and assumptions to be used in the process. At the same time, ACER requested that TSOs also submit the results of LMP (locational marginal pricing) simulations at this stage in order to decide on alternative bidding zone configurations (ACER, 2024a; ACER, 2024b). In August 2022, a second decision on alternative BZ configurations was published by ACER. These were possible solutions to be considered in continental Europe (Germany, France, Italy and the Netherlands) and the Nordic area (Sweden) to improve market efficiency, as can be seen in Table 1. On December 22, 2023, a third decision was made, following the communication of LMP results concerning the Baltic region, which had been absent from the second publication. According to ACER Decision No. 17/2023, 2023, an in-depth and iterative analysis was conducted to evaluate alternative bidding zone configurations for the united synchronous area planned for 2025.

For the Baltic region, multiple alternatives were thoroughly examined. The first proposal involved the merger of the Latvian and Lithuanian bidding zones, or alternatively, the merger of the Estonian, Latvian, and Lithuanian zones, with the aim of reducing price fluctuations and increasing market liquidity. The second alternative proposed splitting the existing bidding zones within individual countries to better reflect internal congestion and improve locational price signals. In addition, as can be seen from Table 1, for some countries multiple configurations are evaluated to balance the different objectives present in market design, such as improving market efficiency, reducing price volatility, ensure network stability and optimize the use of transmission capacity.

From the economic benefits estimated with ENTSO-E projections for the 2025 target year, we can observe that the status quo remains the most efficient solution for most of the countries examined. Only for Germany-Luxembourg do alternative zonal configurations correspond to significant economic benefits, with the division into five zones showing the highest benefits. The detailed results show that an increase in the number of zones does not necessarily correspond to an improvement of the economic outcomes (with 3 zones having lower benefits than 2).

At this stage of the process, Member States have six months to consider the recommendations put forward by the TSOs on ACER's proposed configurations. When the Member States are not unanimous, the European Commission, after consultation with ACER, then also has an additional six months to arrive at a final decision. This process reveals the complexity involved in adjusting bidding zones, both geographically and with reference to the number of zones. Interestingly, while ENTSO-E, ACER, and national TSOs can put forward proposals with respect to configurations of zones, this multiplicity of sources makes the process potentially lengthy and complicated. For example, in the Italian case of introducing Calabria as an additional zone in 2021, the process was relatively swift because it was initiated internally by the national TSO. This demonstrates that, if the national authority opts to reform the market structure, then the process becomes streamlined. By contrast, the lengthy ACER led process will have taken 8 years to recommend a reconfiguration of the German system, which will, quite possibly, be rejected by the German government. This shows the limits of re-zoning led by the EU. The CACM Regulation (Article 34) states that ACER should assess the efficiency of current bidding zone configuration every three years, however the 2018 review will likely complete only in 2026. This slowness reflects the broader challenge of consensus building and harmonization of national and European interests within the redesign of the electricity markets.

5. Examples of zonal pricing systems

This paper aims to address a question frequently asked in both political and academic discussions on electricity market design: what is the appropriate number of zones for an electricity market? In practice, there is no universal answer. The optimal number of bidding zones depends on the context and is determined by a combination of technical, economic, and geographical factors.

For analytical clarity, the analysis concentrates on markets that have progressively increased their number of BZs over time, to examine the dynamics of zone reconfiguration and their implications for market efficiency. Moreover, both zonal and nodal structures allow for hybrid implementations that can enhance system efficiency. The zonal system is always applied on the supply side but not necessarily on the demand side. On the other hand, some nodal pricing systems are applied only on the supply side, such as in California, where retail consumers are subject to a price calculated as the weighted average of the minimum market prices within three load zones, corresponding to the territories of the distribution network operators (DNOs). These examples illustrate how both zonal and nodal pricing systems can be flexibly adapted through

Table 1
ACER Recommendations on Bidding Zones: Country-wise Overview for 2025 target year.

Country	Status Quo	Configurations considered by ACER[b]	N. of Zones	Monetised benefits calculated by ENTSO-E (€ million)[a]
Nordic Europe				
Sweden	4	Config 8	3	−7
		Config 9	3	−34.8
		Config 10	4	−2.2
		Config 11	4	−15.9
Central Europe				
France	1	FR3	3	−9
Germany-Luxembourg	1	DE2	2	264
		DE3	3	251
		DE4	4	312
		DE5	5	339
Italy	7	IT2	8	−60
Netherlands	1	NL2	2	9

[a] The data provided in this table were sourced from Table 53 (p. 105) and Table 62 (p. 127) of the ENTSO-E Main Report on the Bidding Zone Review for the 2025 Target Year.
[b] The names of the proposals are decided by ACER.
Source: Authors' elaboration on ACER decision November 2022 and ENTSO-E Main Report (2025).

Table 2
Number of market zones by zonal pricing market.

Country	Date	Peak demand	Number of zones[a]
Norway	2008 – ongoing	25.2 GW	5
Sweden	2011 – ongoing	25.2 GW	4
Italy	2004 – ongoing	49.6 GW	7[b]
Denmark	2000 – ongoing	6.4 GW	2
Australia	1998 – ongoing	32.8 GW	5
California	1998–2009	43.9 GW	2[c]
Texas	2002–2010	85.5 GW	5[d]
Great Britain	*From 202X?*	48 GW	*7–12?*

[a] The number of zones refers to zonal pricing configurations in place during the indicated periods.
[b] Unlike other systems applying zonal prices to both supply and demand, the Italian market applied zonal pricing only to producers until January 2025. A transition toward a fully zonal system is planned in the coming years, but for now consumers still pay a uniform benchmark price (PUN-Index), a weighted average of zonal prices.
[c] Under nodal pricing, retail consumers are typically settled based on the weighted average of LMPs within three designated load zones, which align with the territories of distribution system operators.
[d] Under nodal pricing, both generators and consumers are exposed to nodal prices.
Source: Authors' elaboration.

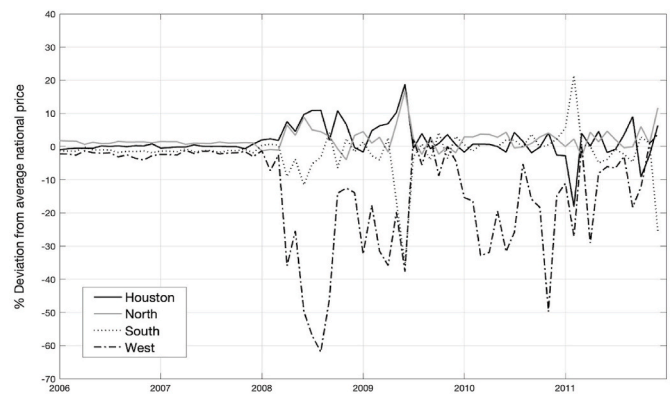

Fig. 2. Percentage deviation of monthly average electricity prices from national average price by bidding zone in ERCOT (2006–2011)
Note: In the absence of an official ERCOT price publicly available on a monthly or daily basis,
we approximate the national average by taking a load-weighted average of the zonal prices (using annual zonal load data).
Source: Author's elaboration based on ERCOT data.

hybrid configurations, enabling policymakers and regulators to balance market efficiency, equity, and political acceptability according to the specific conditions of the system and social objectives.

This comparative perspective is particularly relevant for markets—like Great Britain— that have been considering adopting a zonal design. In recent years, the UK Government has been taking into consideration reforming the energy market, through a wholesale locational pricing design through the consultations DESNZ (2022) and DESNZ (2024). Respectively, in 2023 and in 2025, two FTI Consulting reports[10] have presented different assessments regarding the optimal number of bidding zones, proposing respectively 7 and 12 zones. Recently, the UK government announced that it would not increase the number of GB bidding zones and consider reform of national pricing instead (DESNZ, 2025). This paper offers considerations on the zonal system for markets which still have a national process structure. It is important to note how political and institutional factors have played a decisive role in preserving the status quo, showing that effective market design is not only a technical issue but also related to political economy considerations.

Table 2 offers a brief overview which highlights a key lesson: there is no one-size-fits-all number of market zones, but rather a need for an evidence-based assessment of system needs and market conditions. In this light, the experience of other countries that have implemented—or opted against—zonal reforms offer valuable insights.

5.1. *The case of Texas*

From 2002 to 2010, Texas adopted a zonal market structure, divided into five price zones, each of which had its own electricity price determined by the balance between supply and demand. These separate zones allowed ERCOT[11] to manage congestion, improving reliability and

[10] FTI Consulting (2023) and FTI Consulting (2025).

[11] Electric Reliability Council of Texas.

reducing the risk of blackouts. To monitor inter-zonal congestion, Commercially Significant Constraints (CSCs) were adopted, which represented the operational limits between zones and helped to optimally manage power flow (Mickey, 2018).

The Texas electricity market is an example of how zones can change. ERCOT's State of the Market Reports show that in the early years the market was divided into four zones and then moved to five. In 2007 the market returned to the original four-zone division[12] until its reconfiguration into a nodal structure.

Fig. 2 shows the monthly percentage deviation of the zonal prices from the national average price from 2006 to 2011, during the zonal market structure. Since 2008, the year after the fifth zone North-East was removed, price differences between zones have become significant, showing different trends. In particular, the analysis indicates that all other zones exhibit sustained positive deviations from the national average price, while the West zone shows a trend largely below the national average price. This result may lead to assume that changing the market structure, reducing the number of zones, negatively affected the convergence of zonal prices in Texas.

Subsequently, as of December 2010, the market switched to a nodal system. An interesting aspect to analyze is the difference between nodal and zonal prices. The implementation of the nodal market resulted in lower price volatility than in the zonal market. In these terms, the West zone has continued to have higher levels of volatility than the other zones, which is to be expected given the very high penetration of variable wind generation in that area, as explained by ERCOT (2011). Moreover, in addition to the impact on prices, the switch to the nodal system has been fully implemented both on the demand side and on the supply side, unlike other systems. Within ERCOT, all loads in areas where electricity is open to retail competition are served at nodal prices, which represent about 75 % of customers, exposing them almost entirely to the nodal system (Arup and Limited, 2024).

5.2. The case of California

The Californian electricity market is a second example of the traditional shift from zonal to nodal system. Before 1998, California's transmission network was managed by vertically integrated utilities operating independently. With electricity market deregulation, the California Independent System Operator (CAISO) was created to oversee the high-voltage grid and wholesale market operations statewide (CAISO, 1998a). CAISO adopted a zonal market structure, defined by four BZs[13] Thus, in practice there were actually only two active congestion zones in the ISO system.[14]

It was soon apparent that the zonal system had structural vulnerabilities (CAISO, 2001). Treating intra-zonal congestion as negligible often resulted in unfeasible dispatch schedules, causing considerable stress on real-time operations. This inefficiency manifested itself with particular force during the California energy crisis of 2000–2001, when market instability was caused by the absence of accurate congestion costs and inadequate coordination mechanisms. As a result of these problems, the Federal Energy Regulatory Commission (FERC) issued a series of orders requiring the CAISO to redesign its market and correct its operational deficiencies. The result of these negotiations was the Market Redesign and Technology Upgrade (MRTU), which went into effect on April 1, 2009 (CPUC, 2011). The MRTU replaced the zonal market with a comprehensive nodal system, fundamentally reforming electricity pricing. This transition eliminated the active congestion zones in favor of approximately 3000 nodal price points, enhancing the spatial resolution of price signals.

Despite the implementation of a nodal pricing system in California, consumer exposure remained zonal. CAISO adopts a hybrid model, where consumers are charged the weighted average of nodal prices within three designated load zones, which correspond to the territories of distribution system operators (DSOs). As discussed by Neuhoff (2011), aggregating pricing regions, which involves averaging nodal prices across a region, is a commonly used strategy to mitigate consumer price risk exposure.

Although technically valid, the shift to a zonal system in 1998 received mixed comments, as discussed by Alaywan (2012) in his study about the process of migrating from a decentralized and zonal based market system to centralized and nodal one. Proponents of the zonal, especially some Load-Serving Entities (LSEs) and municipal utilities, system argued that zonal pricing represents a simpler and more transparent market that achieves a better balance between efficiency and fairness and allows greater levels of flexibility and innovation in market engagement. Those who criticized nodal pricing noted that its complexity could minimize transparency and stifle the evolution of personalized market products (Eicke and Schittekatte, 2022). However, supporters of the nodal model, such as CAISO[15] and independent power producers, emphasized its greater economic efficiency, allowing for more efficient dispatching, promoting competitive outcomes and facilitating the optimal location of investments in generation and transmission.

5.3. The case of Australia

The National Electricity Market (NEM) in Australia has adopted a fully zonal structure, composed by five zones along the east coast[16] (Australian Energy Market Commission, 2025). Before the 1990s, Australia's electricity sector was characterized by vertically integrated monopolies confined within state borders. After several challenges in the power generation sector (Simshauser, 2023), the Commonwealth Government initiated a national inquiry through the Productivity Commission, which led to a recommendation for the restructuring and deregulation of the sector. This reform proposal called for a four-zones interconnected grid in 1998. Then, in 2005, Tasmania was connected through an undersea HVDC cable (KPMG, 2013), while the Snowy region was abolished in 2008, showing how BZs structure can vary over time,[17] helping to balance supply and demand more effectively and allowing Tasmania to export excess renewable energy, particularly hydroelectric power, to the mainland.

Moreover, alongside its zonal structure, the Australian electricity market has been also divided into Renewable Energy Zones (REZ), as planned geographical areas where the development of renewable facilities and network upgrades are coordinated to exploit economies of scale and reduce congestion (AEMO, 2018). This framework is a zonal scheme which aims to promote permits and investments in renewables without increasing the number of zonal pricing zones.

The primary goal of the NEM is to integrate the market by balancing

[12] Houston, North, South and West.

[13] North of Path 15 (NP15), South of Path 15 (SP15), Humboldt, and San Francisco. However, due to lack of adequate competition in the Humboldt and San Francisco zones, they had been designated 'inactive zones' and were treated collectively as a single congestion zone included in NP15 (CAISO, 1998b).

[14] These zones, which were later called pricing nodes, were intended to reflect regional transmission limits and take into account the different supply and demand conditions throughout the state, with the intention of simplifying congestion management and providing differentiated price signals for different areas.

[15] Supported by recent evidences provided by CAISO (2023).

[16] Queensland, New South Wales (including the Australian Capital Territory, ACT), Victoria, Tasmania and South Australia.

[17] The zones are defined in accordance with the National Electricity Rules (NER) and any change requires a proposal to amend the rules (Clause 8.9) to the Australian Energy Market Commission (AEMC). However, despite the independent nature of the Australian electricity market, unlike the European one, the individual zones cannot divide themselves internally.

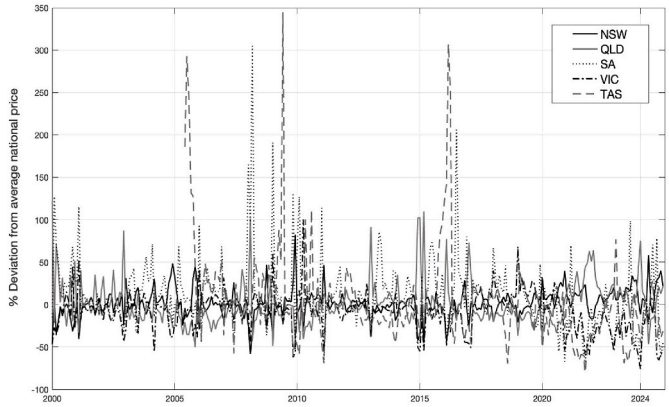

Fig. 3. Percentage deviation of monthly average electricity prices from national average price by bidding zone in NEM (2000–2024)
Note: in order to observe clearly the percentage deviation, the graph does not consider the first price in Tasmania registered in May 2005 (306,87$/MWh).
Note: In the absence of an official national price, we approximate the national average by taking a load-weighted average of the zonal prices.
Source: Authors' elaboration based on AEMO data.

electricity supply and demand, which in turn establishes the price for each zone. Simultaneously, the transmission system employs physical interconnectors to link regions with lower prices to those with higher prices, taking into account issues such as congestion and peak transmission times (Naeem et al., 2022). Fig. 3 shows that the percentage deviations of zonal prices from the national average price are significant throughout the period under analysis, particularly in Tasmania.

This division leads to price discrepancies between markets and worsens reliability issues for regional utilities (ACCC, 2000; IEA, 2001). These limitations result in the regional spot markets not being fully integrated. Typically, electricity flows from lower-price regions to higher-price regions when interconnectors operate at full capacity, but consumer prices still differ between regions. The use of interconnectors to import electricity acts as a buffer for local regions, which cannot quickly expand generation capacity (AEMC, 2020).

The Australian market was among the first to adopt a zonal pricing framework—along with several US states—but, unlike jurisdictions such as Texas and California, chose not to increase price granularity. In ERCOT and CAISO, increased congestion and redispatch costs with zonal pricing, combined with opportunistic bidding behaviour, drove the transition to nodal pricing. The Australian market, on the other hand, retained the zonal structure, considering zonal price signals appropriate. Following the AEMC's Final Report on Transmission Access Reform (2025), all proposals to introduce nodal or hybrid mechanisms were formally withdrawn as the expected benefits did not outweigh the significant implementation costs. Furthermore, Australia, being the sixth largest country in the world by area, has never adopted a nodal pricing model for reasons similar to those that make the implementation of nodal sub-markets in EU difficult: the geographical vastness and the complexity of the nodal system. While ERCOT and CAISO operate over much smaller geographic areas, where it is easier to collect network data at the individual node level, NEM has preferred a zonal approach—less granular but much simpler to manage—coupled with constraint-based pricing mechanisms to deal with network constraints.

5.4. The case of Norway

Turning to Europe, we start with the case of Norway—a country dominated by hydropower as the primary source of electricity generation and whose electricity market is organized into five BZs[18] (Statnett, 2025).

Norway started to operate with flexible and temporary zones in 1991, after the 1990 Energy Act,[19] but only in 2008 its market structure began to transition toward a more formal zonal pricing system, progressively increasing the number of BZs to better reflect transmission constraints and regional grid conditions[20]. Following several zonal reconfigurations, in September 2015 a five-zone structure was consolidated, aligning zonal boundaries with persistent structural bottlenecks in the transmission network. Norway is a clear example of zoning reconfiguration over time, having changed its bidding-zone structure eight times since 2000. This dynamic approach has been important to ensure efficient utilization of the hydro system and transmission network. Based on the report provided by Energy Norway (2023), not only the split but also merger of zones even across national borders should be addressed. If there is no frequent structural congestion between them and price differences are small, countertrading cost would probably be smaller than the benefits a bigger zone would entail for market development, as the Norwegian case shows.

Despite its overall efficiency, the Norwegian zonal market model presents structural limitations. The relatively small number of Elspot[21] bidding zones—the day-ahead market operated by Nord Pool—does not fully reflect underlying grid constraints or provide accurate locational price signals (NVE, 2016). These inefficiencies necessitate the Norwegian TSO Statnett's interventions through redispatch and countertrading, signaling a persistent misalignment between market prices and real-time grid constraints. Ideally, price signals should internalize both marginal production costs and physical transmission limitations. In practice, however, the current model offers distorted incentives for short-term dispatch and long-term investment, hindering optimal infrastructure development and weakening the market's capacity to signal localized scarcity. Nevertheless, the zonal model has been instrumental in managing Norway's hydropower-based system, characterized by high variability due to meteorological and hydrological conditions. Price differentials across zones play a key role in balancing local supply-demand mismatches and promoting efficient resource allocation. Still, persistent remain unaddressed, pointing to the need for continued investment in grid infrastructure.

The Norwegian experience illustrates the trade-offs of zonal pricing, balancing operational flexibility with market coordination, but also revealing the limitations of coarse spatial granularity in renewable-intensive systems within broader European frameworks. In the Norwegian hydropower-dominated system, annual production depends on reservoir levels and inflows, making price signals clear and crucial for scarcity management. The introduction of supply zones has incentivized imports and the prudent use of hydropower during periods of potential shortage. Zonal prices also helped to avoid local crises: in the winter of 2010–2011, higher prices in Central Norway, caused by low reservoir levels and limited imports, stimulated local generation and demand response. Similarly, in spring 2013, price signals in Western Norway

[18] Eastern Norway (NO1), Southern Norway (NO2), Central Norway (NO3), Northern Norway (NO4), and Western Norway (NO5).

[19] The Act covers the generation, conversion, transmission, trading, distribution and use of energy in Norway.

[20] Norway's mountainous terrain and deep coastal waters make infrastructure costly, with major transport flows often routed through Sweden.

[21] Elspot is the day-ahead electricity market used in the Nordic region. Operated by Nord Pool, it serves as an organized trading platform where market participants submit bids to buy or sell electricity for delivery on the following day.

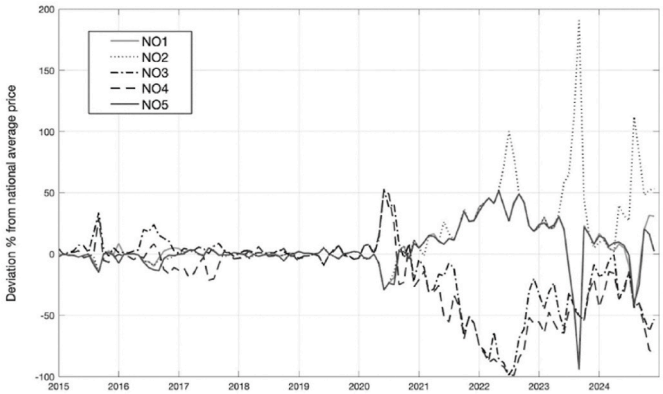

Fig. 4. Percentage deviation of monthly average electricity prices from national average price by bidding zone in Norway (2015–2024)
Note: In the absence of an official national price, we approximate the national average by taking a load-weighted average of the zonal prices.
Source: Authors' elaboration based on ENTSO-E data.

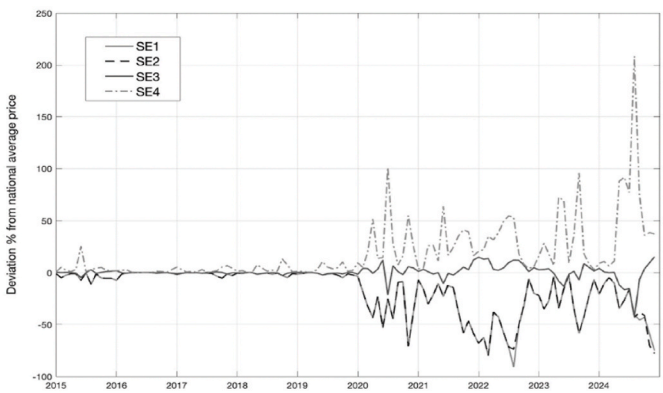

Fig. 5. Percentage deviation of monthly average electricity prices from national average price by bidding zone in Sweden (2015–2024)
Note: In the absence of an official national price, we approximate the national average by taking a load-weighted average of the zonal prices.
Source: Authors' elaboration based on ENTSO-E data.

regulated consumption and generation during maintenance and low inflows, mitigating the risk of shortages (Statnett, 2023b).

Over the past ten years, the Nordic electricity market has experienced a steady increase in traded volumes. Price differences between areas in Norway have generally increased, especially between southern zones and northern zones (see Fig. 4), and deviated from the average during periods of particularly divergent hydrological conditions. Whereas, based on Statnett (2023b), the configuration of the supply zones contributed to increased welfare by providing appropriate incentives to producers and maximizing flows between surplus and shortage areas.

Based on the principle of periodic reconfiguration, in 2018 Statnett has published an analysis which indicates that congestion from the northern supply zone, NO4, is expected to increase (Statnett, 2023a). A split of NO4 has therefore been proposed to manage bottlenecks more efficiently. From NO4 there are connections to central Norway and northern Sweden. However, the skewed loading of the corridors prevents full utilization. Statnett's simulations show that in many cases the line to SE1 fills up first, leaving capacity to NO3 underused. To resolve this, a more granular zonal structure may be required to maintain safe operational limits, splitting northern and southern NO4, which is too large geographically to manage flows safely under the current configuration.

5.5. *The case of Sweden*

Before November 1, 2011, the Swedish electricity market operated as a single bidding zone, with prices set on the Nord Pool exchange. Electricity was sold from the North, where low-cost hydropower is concentrated, to the demand-heavy South. However, persistent bottlenecks in north-south transmission meant that contracted volumes could not always be physically delivered. In such cases, Svenska kraftnät, the Swedish TSO, generally reduced exports, particularly to continental Europe, to uphold a uniform national price despite internal congestion.[22] In the early 2000s, Svenska kraftnät reduced interconnection capacity during internal congestion, effectively prioritizing domestic over cross-border transmission. This practice led to discriminatory treatment of customers by location, undermining market integration within the EEA Agreement. Consequently, the European Commission urged Sweden to stop export curtailments on the spot market, reporting it as a distortion of competition (European Commission, 2010).

In 2007, the Swedish Energy Markets Inspectorate, the TSO, Swedenergy (an umbrella group for producers), and the Confederation of Swedish Enterprise jointly published a report (POMPE, 2007), examining how to split the market. Known by the Swedish acronym POMPE, the report recommended dividing Sweden into two separate price zones. Following several years of inquiry, in 2010 Sweden issued a decision mandating that Sweden be split into distinct price zones by 2012. This episode highlights how external institutional pressure—in this case, from the European Commission—can act as a catalyst for market reconfiguration[23](European Council, 2010). To comply with EU regulations, Sweden adopted a market-splitting approach, introducing four bidding zones.[24] The goal of this restructuring was to better reflect physical transmission constraints and provide more efficient price signals, encouraging targeted investments to alleviate congestion (Lundin, 2021).

Holmberg and Tangerås (2023) show that one of the most relevant impacts of zonal pricing is a significant price divergence between the southern zone SE4 and the rest of Sweden.

(Fig. 5). SE4 experiences higher prices due to limited transmission capacity from the northern regions and Denmark. In the first months following implementation, price differences were particularly pronounced due to cold winter temperatures that increased demand, reduced transmission capacity between Sweden and Finland that limited energy imports, and temporary outages at Ringhals nuclear reactors that constrained supply. As expected, over time SE4 prices became more aligned with those in DK2 (Western Denmark) due to geographic proximity and strong electricity trade between the two areas, showing consistent price differences across various zones compared to the average national price.

At the same time, in the long term, the new zonal configuration

[22] This practice was one of the exemptions granted by ACER, allowing member states to limit commercial flows by up to 30 % under the 70 % rule for congestion management (Holmberg, 2024; ACER, 2023).

[23] This practice was encouraged by the European Commission with the Electricity Regulation (EU) 2019/943, which requires TSOs to allocate transmission capacity on a non-discriminatory basis. In the case of Germany, however, the situation is more complex. Although Germany has not introduced multiple bidding zones, it has reached an agreement with Denmark to comply with the 70 % rule in the day-ahead market, while addressing internal congestion through extensive redispatching and counter-trading costs for which Germany bears full responsibility, spending several billion euros annually on counter trading. In contrast, countries like Norway, outside the EU framework and not bound by the 70 % rule, typically manage congestion by reducing exports. This reflects the approach taken by other EU countries prior to the implementation of stricter rules on cross-border trade.

[24] SE1 (Luleå), SE2 (Sundsvall), SE3 (Stockholm), and SE4 (Malmö).

brought structural benefits, including increased competition, transparency, and clearer investment signals for strengthening the transmission network and adding new generation capacity in critical areas. It also underscored the need for further infrastructure upgrades to relieve congestion between zones. Overall, the reform marked a shift toward a more efficiency-driven market, aligning Sweden with European policies on market integration and cross-border electricity trade.

To observe the effects of Sweden's zonal market division, we refer also to Loiacono et al. (2025). The aim of their work is to analyze the impact of the shift to the zonal system in Sweden on electricity prices. Their findings show that SE1 and SE2 in the North have a positive demand-supply balance, while SE3 and SE4 in the South face higher prices and a negative balance.[25] In particular, in the northern zones (SE1 and SE2) the price increase is similar, while in SE3 and SE4 the effects are more pronounced. This indicates that zoning has indeed differentiated territorial prices, especially in the South, where demand and population concentration are higher and there are transmission constraints from the North. Moreover, the overall price of energy increased across Sweden after the implementation of zonal pricing in the next 20 days after the reconfiguration.

The initial increase in zonal prices following the splitting of the zones in 2011 was particularly evident in the first 20 days.[26] However, when examining the price trend over the following two years, a reduction can be observed compared to the uniform price calculated in the two years prior to the reconfiguration.[27] This phenomenon can be interpreted as a typical learning period in electricity markets, during which the market adjusts to the new zone configuration (Doraszelski et al., 2018). In fact, within the framework of its Bidding Zone Review Process, ACER generally includes a transition period during which stakeholders assess the impacts on liquidity, congestion, price convergence, and overall market efficiency. This monitoring period, which may last several months, is crucial for gathering data and providing assessments after the initial adjustment phase.

An interesting point not directly studied by these authors would be to observe, through a counterfactual analysis, how the energy price in Sweden would have been in the absence of the market reconfiguration. However, it can be seen that domestic demand and imports remained mostly unchanged in the two years before and after the implementation in November 2011 (respectively a change of 1.16 % and 1.18 %), while exports increased considerably, with a change of 155.41 % before and after treatment, considering a time interval of four years (two years before reconfiguration and two years thereafter). This corresponds to what we hypothesize as a possible explanatory scenario: exports rose after Sweden was no longer permitted to limit them, which contributed to higher prices in SE4. These exports were sensitive to the emerging price differentials revealed by the zonal structure, and the new incentives led to increased flows toward Continental Europe. As export volumes increased, transmission constraints on the southern interconnectors became increasingly tight, operating at or near capacity. According to our interpretation, this export constraint contributed to the rise in electricity prices in SE4, as surplus local supply could no longer flow outward to meet external demand, thereby pushing prices upward to clear the local market. From this perspective, the price increase observed in southern Sweden after the reconfiguration would not primarily reflect changes in national-level supply and demand, but rather the effect of structural bottlenecks and enhanced export incentives brought about by the new zonal market design.

5.6. The case of Italy

Before 1999, electricity trade in Italy consisted of a patchwork of bilateral agreements within regional monopolies. The market reform of March 16, 1999 established a centralized wholesale market, opening the door to transparent and coordinated trading and the first taste of congestion management. With the launch of the Italian Power Exchange (IPEX) in 2004, spot prices were determined by supply and demand within each geographical zone, constrained by physical transmission limits and complemented by 'virtual' zones for international interconnectors and limited production hubs, as Brindisi and Rossano. Since 2021, the Italian electricity market is constituted by seven zones.[28]

The increase in Renewable Energy Systems (RESs) capacity and the resulting changes in generation profiles have significantly altered power flow patterns and shifted the location of structural congestion. These changes lead to higher re-dispatching costs for TSOs and a reduction in social welfare, prompting the redefinition of bidding zones as a key solution (Terna, 2024), as happened in Italy. Before December 31, 2020, the Italian electricity market was divided into six bidding zones. However, since January 2021, the configuration of BZs has been modified. Firstly, the Umbria region was reassigned from the Centre-North to the Centre-South zone. This adjustment aimed to better account for regional negotiations and congestion impacts, ensuring that the zonal structure aligned with the geographical distribution of generation and demand and the operational dynamics of the electricity market.[29] A significant update was the creation of the Calabria bidding zone, introduced to better manage renewable generation growth in the region. By treating Calabria as a separate zone, the market could capture price signals more accurately and improve power flow management.

This reconfiguration has been based on expert assessments of ENTSO-E (2023b), raising questions as to whether these changes will help improve market liquidity and efficiency, in order to adapt the market structure to the changing energy landscape. As happened in Italy, the decision was politically accepted and facilitated when changes in the area are not expected to significantly alter average electricity prices. In the Italian market, as zonal prices were very similar in recent years, this reflected a wider dynamic where alignment with EU recommendations was more likely as domestic economic impacts were considered to be limited. On January 1, 2025, the decision was taken to remove the hybrid system so far adopted, whereby consumers paid a single national price (PUN), a demand-weighted average of zonal prices, shifting all end customers to their local zonal price[30]. The PUN now serves as a benchmark index, which is expected to be gradually phased out in the coming years.

Electricity zone prices in Italy show a trend of gradual convergence, distinguishing themselves from countries such as Sweden and Norway, where price differences between zones remain more marked and persistent. In Italy, the differentials between supply zones have narrowed over time, a sign that investments in network capacity, together with more efficient market dynamics, are helping to equalize price signals. This harmonization process reflects a greater integration of the internal market and a reduction in structural congestion, resulting in a flattening of zonal spreads. Fig. 6 shows the percentage deviation of the monthly zonal prices of each zone of the Italian electricity market from 2015 to 2024 (considering Calabria from 2021 onwards) compared to

[25] See Table A.1, p. 8 in Loiacono et al. (2025).
[26] See Figure 2, p. 4 in Loiacono et al. (2025).
[27] See Figure A.3, p. 8 in Loiacono et al. (2025).
[28] North, Centre-North, Sardinia, Centre-South, South, Calabria and Sicily.
[29] It should be noted that, in the case of Umbria, this change was primarily a boundary movement, meaning that the adjustment mainly involved redefining the regional boundaries to better align with actual grid dynamics rather than representing a wholesale change in market conditions.
[30] In some cases, applying uniform consumer prices across areas where congestion occurs can mute locational signals, preventing demand from adapting to local supply conditions and potentially aggravating network constraints. As noted by Neuhoff et al. (2025), allocating a fixed amount of Financial Transmission Rights (FTRs) to vulnerable consumers could mitigate these effects by protecting them from price volatility while preserving efficient locational incentives.

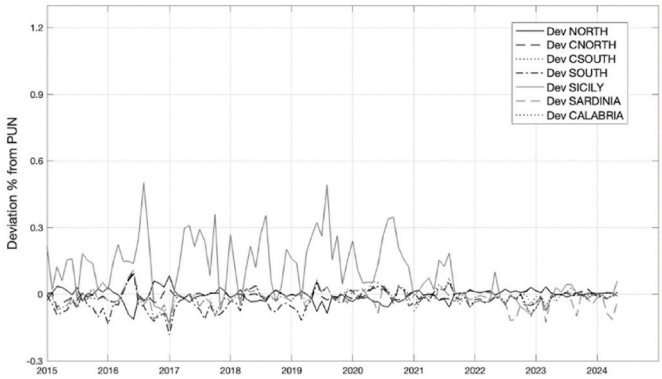

Fig. 6. Percentage deviation of monthly average electricity prices from PUN by bidding zone in Italy (2015–2024)
Note: In the Italian case, we use as reference national price, the PUN, the average price of electricity purchased on the Italian market, calculated daily by the Gestore dei Mercati Energetici.
Source: Authors' elaboration based on GME data.

the national average price (PUN), from which it can be seen that the islands in particular have had greater divergence from the total average of prices. Note that the left scale in Fig. 6 is much smaller than both Figs. 4 and 5.

The process of reconfiguration for the Italian market started in 2019 by the Italian TSO, Terna, which has developed a number of alternative configurations based on a specific, so-called 'Expert-based' approach, whereby TSOs determine the zonal structure by leveraging statistical analyses, their in-depth knowledge of the electricity system, and insights from various relevant studies in the *Annex 4: Justification of configurations of the Bidding zone review region Central Southern Italy* (ENTSO-E, 2019). This study includes the Ten Year Network Development Plan,[31] the Medium-Term Adequacy Forecast,[32] the National Development Plan, and other technical assessments conducted by Terna. While such expert-based methods have proven effective in Italy, alternative "model-based" approaches employing clustering algorithms could offer more optimized zonal delineations. However, these remain under development and require further refinement. In its latest review, Terna maintained the expert-driven approach, while planning a gradual integration of model-based techniques, evaluating six alternative BZ configurations (ENTSO-E, 2023a).

Another proposal was presented by the study conducted by ENTSO-E (2024a) on the impact of the configuration of BZs on market liquidity and transaction costs. ENTSO-E evaluates the potential division of the North zone into two new areas: a large eastern zone (ITI1) and a smaller western zone (ITI2). The division leads to a marked decline in generation and demand volumes in ITI2—approximately 80 % compared to the status quo—while ITI1 experiences only marginal reductions. Despite these changes, market concentration remains stable: the Residual Supply Index (RSI)[33] increases in ITI2 and decreases in ITI1, yet remains above the critical threshold in all cases, and the Pivotal Supplier Index (PSI)[34] is consistently zero. This suggests that the expected decline in liquidity, more pronounced in ITI2, is not driven by concentration effects. Price correlation dynamics also shift, with ITI1 showing improved price alignment and ITI2 exhibiting a modest decline. Overall, the analysis underscores that while the reorganization does not raise competition concerns, it may reduce market liquidity and size, particularly in smaller zones, illustrating the complex trade-offs inherent in zonal market restructuring.

6. Lessons learned from the zonal reconfiguration of electricity markets

Based on this comparative analysis, we show that zonal configurations can offer significant advantages in aligning price signals with the physical constraints of transmission networks, while introducing design challenges and trade-offs.

In the European context, Italy stands out as the most successful example of a zonal pricing system: since its introduction, price differentials between zones have remained remarkably low. This convergence largely reflects the high degree of interconnection of the Italian transmission network, including the creation and reinforcement of links with Sicily and Sardinia, which have significantly reduced internal congestion. Another contributing factor is the frequent role of combined-cycle gas turbines (CCGTs) as the marginal technology, a feature that tends to harmonize marginal costs across zones.[35] Moreover, the zonal design has supported targeted investments in renewable generation, particularly in Southern Italy, where prices historically exceeded the national average, promoting convergence among zonal prices and enhancing overall market efficiency. Finally, the 2021 revision of supply zones proved to have improve local price signals, confirming that well-targeted boundary adjustments can improve market efficiency. Even during the Covid-19 and Ukraine crises, Italy's zonal prices fluctuated only marginally around the average, while markets such as Sweden and Norway—without hybrid arrangements or a common purchase price—showed pronounced north-south disparities undermining consumer equity.

By contrast, the process of reviewing European market zones remains cumbersome: the absence of a single decision-making authority and the need for consensus between national TSOs, regulators and member state governments regularly delays or dilutes proposed reconfigurations. However, Australia, despite being a single jurisdiction market, has found substantial price divergence between market zones, but has refrained from further subdivision, pointing out that both the political-regulatory environment and sheer geographic scale limit not only the move to nodal pricing, but also incremental zone refinements. In both cases, the Italian experience suggests that hybrid pricing models – zonal on the supply side and single 'national' price on demand side - may offer an ideal compromise: protecting consumers from excessive location-based price fluctuations while preserving the administrative simplicity of a zonal framework.

Despite the diversity of experiences examined, several general conclusions can be drawn. First, zonal pricing can improve transparency and operational efficiency by signaling congestion and guiding investment. However, the benefits depend on the appropriate definition and

[31] The TYNDP offers a European-level perspective that complements national grid expansion strategies. It outlines a coordinated vision for the future electricity system, assessing how cross-border interconnections and energy storage solutions can facilitate the energy transition efficiently and securely.

[32] The MAF assesses the adequacy of supply to meet demand in the mid-term time horizon while considering the connections between different power systems across the European perimeter.

[33] 'The RSI for a company i measures the percent of supply capacity remaining in the market after subtracting company i's capacity of supply' (Newbery et al., 2004).

[34] 'The PSI examines whether a given generator is necessary (or 'pivotal') in serving demand. In particular, it asks whether the capacity of a generator is larger than the surplus supply (the difference between total supply and demand) in the wholesale market' (Newbery et al., 2004).

[35] ARERA, 2025.

periodic re-evaluation of zone boundaries. Second, while smaller or more granular zones can internalize constraints more effectively, they may also raise concerns over liquidity, particularly in long-term markets. Yet, as shown in Italy and the Nordic region, institutional mechanisms such as market coupling and coordinated planning can mitigate these effects. Third, price volatility and welfare impacts vary by context: while zonal markets may introduce short-term price differentiation, they can also reduce redispatch costs and deliver longer-term efficiency gains. Finally, a persistent gap in literature and policy practice is the lack of empirical evidence on the true welfare effects of bidding zone reconfigurations. Most existing studies are based on simulation models or partial indicators. There remains a need for more systematic and evidence-based assessment of past reforms, including their impact on consumer surpluses, producer behaviour and system-wide efficiency.

In light of the results presented, we suggest a dynamic and evidence-based approach to zonal area design. Rather than considering the number of zones as fixed, regulators should consider zonal boundaries as a policy tool that evolves with network constraints, market developments and decarbonization targets. Responsive zone design, supported by expert and model-based assessments, can balance efficiency, liquidity and equity in increasingly complex and integrated energy systems. However, it is equally important to acknowledge that once a zonal structure is implemented, it tends to exhibit institutional and political persistence. This reflects a fundamental trade-off: while more granular reconfigurations may enhance short-run allocative efficiency by better aligning market outcomes with physical grid constraints, frequent structural adjustments can introduce regulatory uncertainty, undermine investment signals, and disrupt long-term market stability. As such, zonal design should be approached with a clear understanding that reconfiguration entails both technical and institutional costs, and that the benefits of greater operational precision must be weighed against the systemic value of predictability and continuity.

CRediT authorship contribution statement

Marta Moretto: Writing – review & editing, Writing – original draft, Supervision. **Michael G. Pollitt:** Writing – review & editing, Writing – original draft.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Marta Moretto reports financial support was provided by SDA Bocconi School of Management. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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